English Medium

EDU TERIA

72nd BPSC

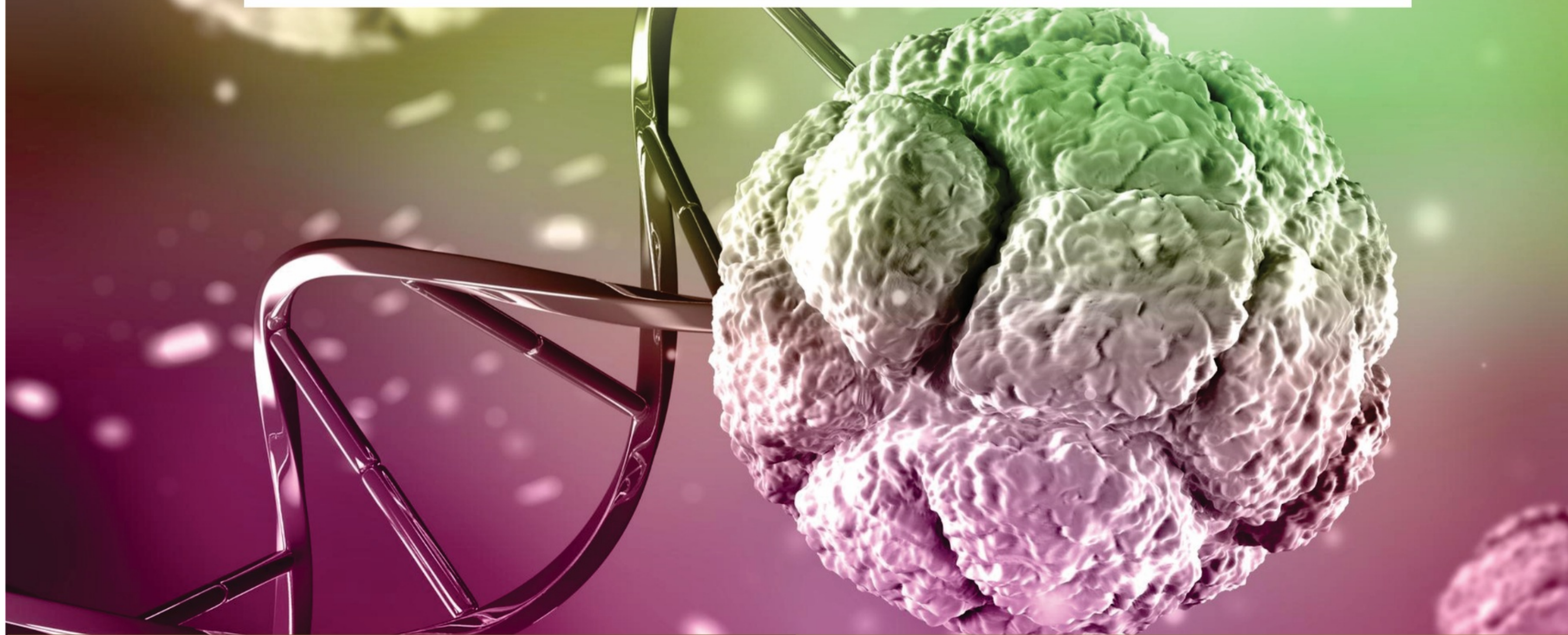
Prelims Online Batch

BIOLOGY

Topic wise **Notes**

LECTURE 20

Excretory System & Reproductive System



Excretory System & Reproductive System

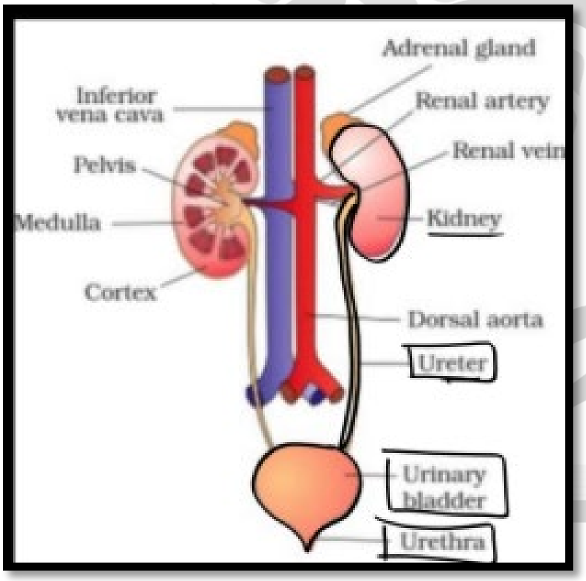
Excretory System

The Human Excretory System is a vital biological system responsible for removing metabolic waste products from the body and maintaining internal balance (homeostasis). It primarily deals with the elimination of nitrogenous wastes formed during metabolism

- Every living organism generates waste in its body and has a mechanism to expel it.
- The waste products that are produced are urea, uric acid, ammonia, carbon dioxide, water and ions like sodium, potassium, chloride, phosphates etc.
- Ammonia, urea and uric acids are the major sources of "nitrogenous wastes".

Modes of Excretion in Animals

- Modes of Excretion in Animals Different types of animals have a different mechanism for excretion. This depends upon the environment, habitat and daily water intake.



Nitrogenous Wastes

Nitrogenous wastes are toxic byproducts produced by breaking down proteins and nucleic acids, requiring excretion to avoid poisoning.

- Nitrogenous waste is synthesized by animals from excess amino acids.
- Unlike lipids and carbohydrates amino acids are not stored in the body.
- They are deaminated by the liver tissues, and the amino group has to be excreted.

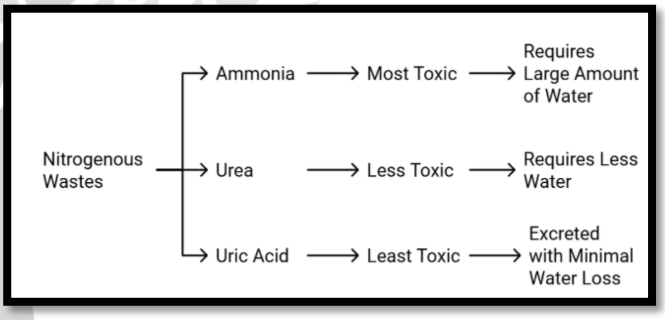
Types of Nitrogenous Wastes

1. Ammonia (NH_3)

- Highly toxic
- Requires a large amount of water for excretion
- Highly soluble in water

Excreted by:

- Aquatic animals (e.g., fish, amphibian larvae)
- Process:
- Directly diffuses into surrounding water
- Animals excreting ammonia are called Ammoniotelic



2. Urea

- Less toxic than ammonia
- Requires moderate amount of water
- Formed in the liver via urea cycle

Excreted by:

- Humans, mammals, amphibians (adult stage)
- Animals excreting urea are called Ureotelic

3. Uric Acid

- Least toxic
- Almost insoluble in water
- Excreted as semi-solid paste (saves water)

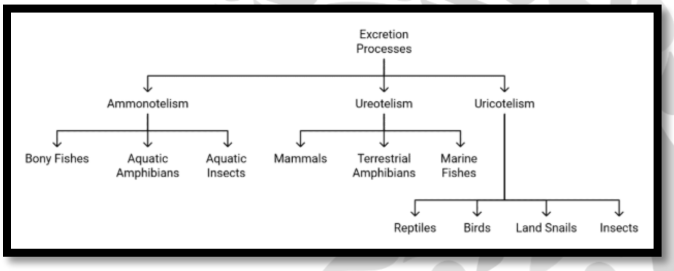
Excreted by:

- Birds, reptiles, insects
- Animals excreting uric acid are called Uricotelic

Category	Product formed	Solubility in water	Examples
Ammonotelic	Ammonia (highly toxic)	Highly soluble, therefore needs plenty of water for its excretion.	Fresh water aquatic animals e.g. bony fish, *Amoeba*
Ureotelic	Urea (less toxic)	Less soluble, thus needs less water for excretion	Mammals like humans, dog etc, marine fishes and amphibians like frog and toad
Uricotelic	Uric acid (least toxic)	Insoluble solids or semi solid. Needs very little water just to flush out the uric acid	Birds, reptiles and insects.

Ammonotelism

- The process of eliminating “ammonia" from the body is known as ammonotelism and the organisms which exhibit this nature are called ammonotelic.
- Example: bony fishes, tadtole larva etc.
- Ammonia is quite soluble, so it is excreted in the form of ammonium ions either by a simple diffusion process across the body surface or through the gills.



Uricotelism

- Elimination of "uric acid" as the main nitrogenous waste material is called uricotelism. Animals who exhibit this kind of elimination are called uricotelic animals.
- Example: Reptiles, birds, snails.
- In some mammals and amphibians, "urea" is excreted and is called ureotelic.
- In these organisms, ammonia that is produced is converted to urea in the "liver" of animals and is released in the blood which is filtered and released out by the kidneys.
- It helps in maintaining the "osmolarity" in the body.

Ureotelism

- It helps in maintaining the "osmolarity" in the body.

  $2NH_3$ (Amonia) + $2CO_2 \rightarrow NH_2\text{-}CO\text{-}NH_2$ (Urea) + H_2O

Organs of Human Excretory System

The organs of the Human Excretory System are responsible for removing metabolic wastes (especially nitrogenous wastes like urea) and maintaining internal balance (homeostasis).

The human excretory system is a well developed system and it includes the following part:.

Skin

- The skin helps in excretion by the way of sweat.
- The skin eliminates compounds like NaCI, some amount of urea etc.

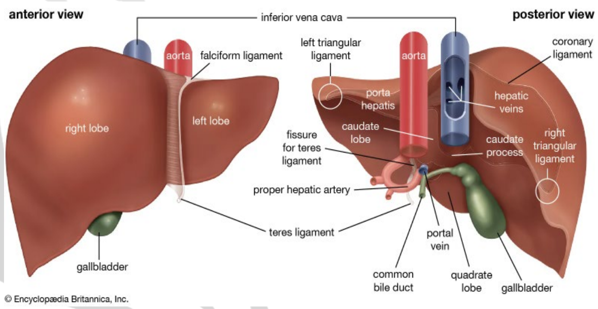
The human excretory system is a well-developed system and it includes the following part:

Lungs

- Lungs are the primary respiratory organs and they help take in oxygen and expel "carbon dioxide (CO_2)”.
- In this process, they also function to eliminate some amount of water in the form of vapour.

Liver

- The liver has an important function in excretion.



- The liver changes the decomposed hemoglobin of the worn-out “red blood corpuscles" into bile pigments "bilirubin" and "biliverdin”.
- These pigments are passed into the alimentary canal with the bile for elimination in the faces.
- The liver also excretes cholesterol, steroid hormones, certain vitamins and drugs through the bile.
- The liver also excretes cholesterol, steroid hormones, certain vitamins and drugs through the bile. The excess amino acids in the body are deaminated by an enzyme

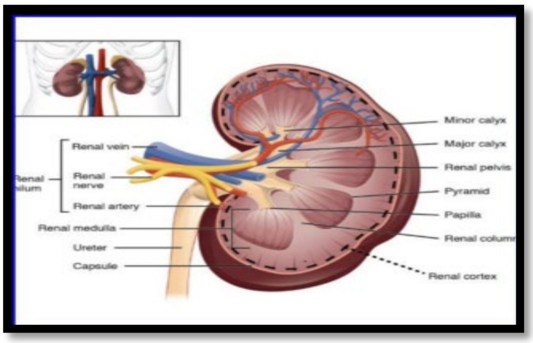
oxidase, producing ammonia NH3. Ammonia being toxic is converted to "Urea"".

Kidneys

- The Kidney is the main organ of the human excretory system.
- The Kidneys are paired organs in each individual. They are the "primary excretory organ" in humans and are located one on each side of the spine at the level of the liver.
- The human excretory system comprises of the following structures:
- 2 Kidneys
- 2 Ureters
- 1 Urinary bladder
- 1 Urethra

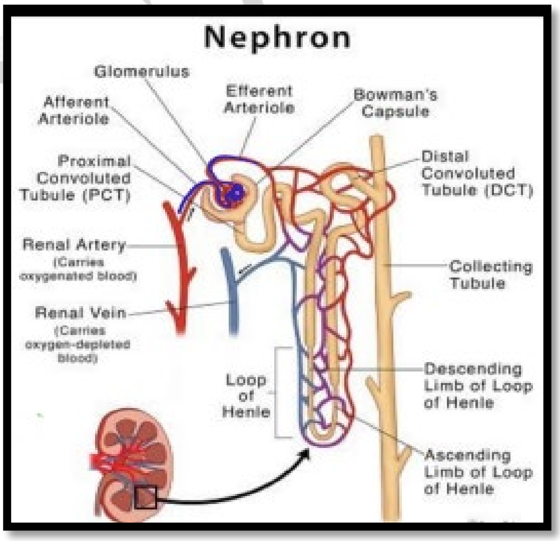
Kidney's Anatomy

- **Renal Capsule:** It is the outer membrane that surrounds the kidney; it is thin but tough and fibrous.
- **Renal Pelvis :** the basin-like area that collects urine from the neurons, it narrows into the upper end of the ureter.
- **Calyx :** It is the extension of the renal pelvis; they channel urine from the pyramids to the renal pelvis.
- **Cortex :** It represents the outer region of the kidney; extensions of the cortical tissue contain about one million blood filtering nephrons".
- **Nephron :** These are the filtration units in the kidneys. The nephron is the "Structural and Functional unit of Kidney".
- The medulla is the inner region of the kidney that contains 8-12 renal pyramids. The pyramids empty into the calyx.
- Medullary pyramids are formed by collecting ducts which are the inner part of the kidney.
- The ureter collects filtrate and urine from the renal pelvis and takes it to the bladder for urination.



Nephron's Anatomy

- **Renal Artery :** It brings waste-filled blood from the aorta to the kidney for filtering in the nephron.
- **Glomerulus Apparatus:** Each glomerulus is a cluster of blood capillaries surrounded by Bowman's capsule. Its main function is "Ultrafiltration".
- The Glomerulus and Bowman's capsule is together called "Renal Capsule".
- Proximal convoluted tube (PCT)
- **Loop of Henle:** Thin descending limb of the loop of Henle and thick Ascending limb of the loop of Henle. The capillaries supplying the blood to
- Loop of Henle is called "Vasa Recta".
- Distal convoluted tubule (DCT)
- **Renal Vein:** When filtration is complete, blood leaves the nephron to join the renal vein, which removes the filtered blood from the kidney
- **Arterioles:** Blood is brought to and carried away from the glomerular capillaries by two very small blood vessels which are the afferent and efferent arterioles.



Functions of Nephron

Renal Capsule

- The blood carried by the afferent arteriole into the glomerulus, the pressure created pushes small molecules through the capillaries and into the glomerular capsule.
- This separates water, ions and small molecules from the blood, filter out wastes and toxins and returns needed molecules to the blood.
- This is called Ultrafiltration.
- This substance, lacking blood cells and large molecules in the bloodstream, is known as an ultrafiltrate.
- The ultrafiltrate travels through the various loops of the nephron, were water Next, the ultrafiltrate must travel through a winding series of tubules.

Proximal Convoluted

- Nearest to the glomerulus.
- They have permeable cell membranes that reabsorb glucose, amino acids, metabolites and electrolytes into nearby capillaries and allow for circulation of water.

Loop of Henle

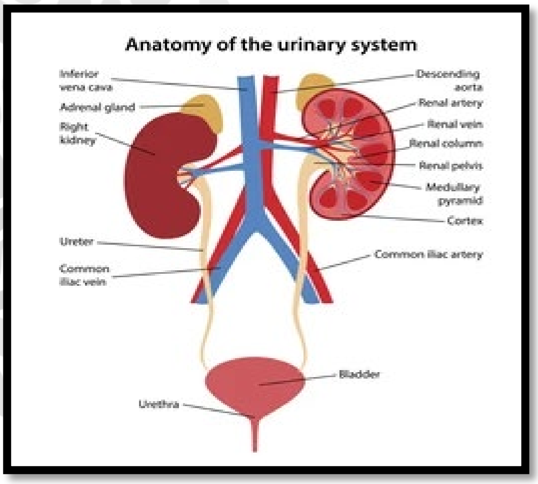
- When the filtration reaches the descending limb of the loop, water content has been reduced by 70%.
- The filtration contains high levels of salt (mostly sodium).

Distal Convoluted Tubule

- The Loop of Henle leads into the distal convoluted tubule which is where the kidney hormones cause their effect.
- And the distal convoluted tubule leads to collecting ducts.
- The collecting ducts together form the renal pelvis through which the urine passes into the ureter and then into the urinary bladder.
- Helps regular potassium excretion.
- The kidneys collect and get rid of waste from the body in 3 steps.
    - **Glomerular filtration:** Filtrate is made as the blood is filtered through a collection of capillaries in the nephron called glomeruli.
- The kidneys collect and get rid of waste from the body in 3 steps Tubular reabsorption:
    - **Tubular Reabsorption :**The tubules in the nephrons reabsorb the filtered blood in nearby blood vessels.
    - **Tubular secretion:** The filtrate passes through the tubules to the collecting ducts and then takes them to the bladder.

Urine and its Components

- Urine is made of water, urea, electrolytes, and other waste products:
    - Some medicines and drugs are excreted in urine and can be found
    - in the urine.
    - 95% water
    - 2% urea



Mechanism of Urine formation

(i) **Ultrafiltration—** Filtration of fluid from glomerular into bowman's capsule known as ultrafiltration.

(ii) **Reabsorption—** Nutrients salts present in glomerular filtrate get reabsorbed by vasa recta (network of efferent arteriole)

(iii) **Tubular Secretion—** Active sensation harmful substances from blood vessel to rend tubule know as.

Urine and its Components

- Urine is made of water, urea, electrolytes, and other waste products.
    - 0.35% sodium
    - 0.6% chloride
    - 0.15% potassium
    - 0.25% phosphate
    - 0.25% Sulphate
    - 0.1% Creatin
    - 0.05% Uric acid

Reproductive System

Groups of organs involved in production of gametes result in formation of new offspring known as Reproductive System.

Reproduction occurs in following stages:

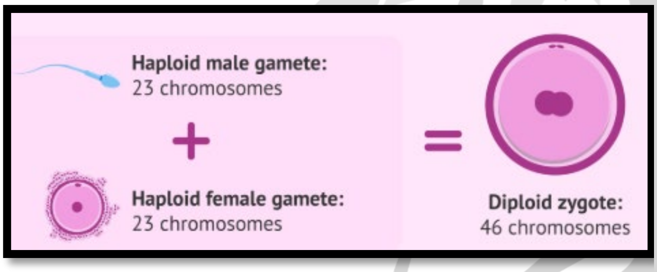
- Gamete formation
- Fertilization
- Pregnancy (Development of foetus)

Male Sex Organs	Female Sex Organs
• Primary sex organ Testes–one pair • Secondary sex organ Beard Mustache	• Primary sex organ One pair ovary • Secondary sex organ Breast & Vagina

Gamete Formation

(i) Male –

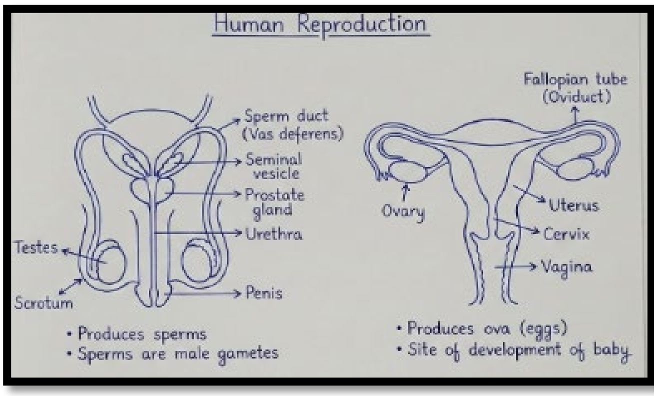
- **Spermatogenesis**– Formation of male gamete in tests.
- Male gamete is known as sperm. Sperms are haploid Cells (n).
- Sperms are formed by the meiosis division.



(ii) Female–

- Gametes are formed with the process known as oogenesis

Human Reproductive System

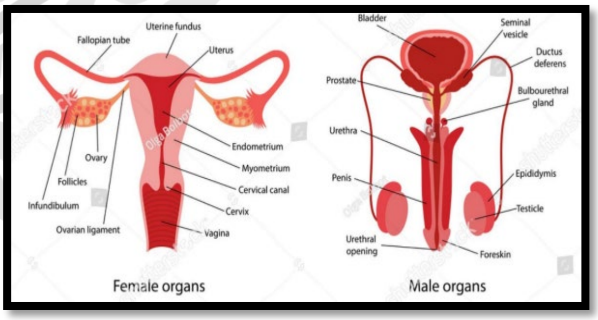


- The human reproductive system is a collection of internal and external organs in males and females that work together to produce offspring through sexual reproduction.

- It involves the production of gametes (sperm and eggs), fertilization, and the nurturing of a fetus, all regulated by hormones to ensure species' survival

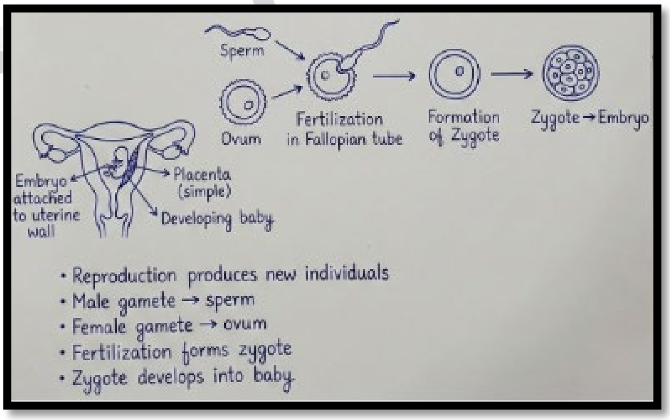
Male Reproductive System: Primarily responsible for producing and delivering sperm (spermatogenesis)

- **Teste:** Produce sperm and testosterone.
- **Scrotum:** Houses testes outside the body for a lower temperature required for sperm production.
- **Epididymis & Accessory Glands:** Store, transport, and nourish sperm.



Female Reproductive System: Produces eggs (ova), supports fertilization, and nurtures fetus

- **Ovaries:** Produce eggs and sex hormones (estrogen/progesterone).
- **Fallopian Tubes:** Transport the egg and are the site of fertilization.



- **Uterus:** A muscular organ that holds and nourishes the developing fetus.

- **Vagina:** The canal for intercourse and childbirth

Reproductive Processes

- **Gametes:** Females produce eggs in the ovaries, and males produce sperm in the testes.
- **Fertilization:** Fusion of sperm and egg occurs to form a zygote, usually in the fallopian tube.
- **Menstrual Cycle:** If no fertilization occurs, the uterine lining sheds (menstruation).
- **Gestation:** The fetus grows within the uterus for roughly nine months.
- **Puberty:** Reproductive organs grow and become active, resulting in the ability to reproduce.